Dr. Ferhat KUTLU
Cognitive AI Scientist

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Personal Statement

Graduation from the National Defense University, system engineering branch, is followed by six years in the army units. The next three years were spent with an M.Sc. at the Bilkent University, Computer Engineering Department. Since then, much knowledge has been gained within tens of ITC and research projects. Parallel to those, a Ph.D. study is accomplished at the Middle East Technical University, Graduate School of Informatics, Cognitive Science Department. Engaged in innovation projects about the areas of Cognitive-AI, Affective Computing, Federated Learning, Image Processing, Video Analytics, 3D, eXtended Reality, and electro-optic communication – targeting pain-points in sustainability, energy efficiency, health informatics, and smart cities.

Core Skills

A balanced blend of technical, academic, and managerial wisdom backed by years of experience to ensure the harmonization of all resources. Inclusive and experiential teaching. Researcher for project ideation, lean on AI and deep learning. Supported by decades of project experience, a competent combination of scientific approaches and contemporary management skills. International consortium building. Agile project management and performance evaluation by using the OKRs.

Education

- ✔ **PhD** in **Cognitive Science**, Graduate School of Informatics, METU (Middle East Technical University), ANKARA/TÜRKİYE, 2012-2023
  Accomplished with the thesis "Identification of Discourse Relations in Turkish Discourse Bank" about Discourse Analysis for Natural Language Understanding – fine-tune LLMs with the modern corpora and ML artefacts.

- ✔ **MSc** in **Software Engineering** (Including BS Completion), Bilkent University, ANKARA/TÜRKİYE, 1998-2001
  Accomplished with the thesis "Categorization in a Hierarchically Structured Text Database" about Data Mining about Text Categorization Algorithms, implemented w/ C++, to build a task-specific decision tree trained on corpora derived from NewsGroups hierarchical data.

- ✔ **Bachelor's** in **System Engineering**, National Defense University, ANKARA/TÜRKİYE, 1988-1992

Publications

Kutlu, F. (2026). **AI Support for Improving Managerial Decision-making Processes Regarding Workforce Planning**. In: Vesa Salminen (eds) Human Factors, Business Management and Society. AHFE (2026) International Conference. AHFE Open Access, vol 215. AHFE International, USA. https://doi.org/10.54941/ahfe1007607

Kutlu, F., Zeyrek, D., & Kurfalı, M., **Toward a shallow discourse parser for Turkish**, Natural Language Engineering, 1-26. doi:10.1017/S1351324923000359, **Cambridge University Press**, 2023.

Ferhat Kutlu and H. A. Güvenir, “Categorization in a Hierarchically Structured Text Database” in: Proceedings of the Tenth Turkish Symposium on Artificial Intelligence and Neural Networks, (TAINN’2001), A. Acan, I. Aybay, and M. Salamah (Eds.), Gazimagusa, Turkish Republic of Northern Cyprus (June 2001), 218–227.

Employment History

Lecturer, **Beykoz University**, İSTANBUL/TÜRKİYE, Sep 2026 – Ongoing
Teach AI and software engineering courses in the AI Department of Engineering Faculty.

Lecturer, **Fenerbahçe University**, İSTANBUL/TÜRKİYE, Aug 2025 – Aug 2026
Teach software engineering courses in the Management Information Systems department of the Faculty of Economics, Administrative and Social Sciences

General Manager, **Bahçeşehir University Innovation and Consultancy Inc.**
İSTANBUL/TÜRKİYE, Nov 2023 – Aug 2025
Co-author the project proposals towards the Ministry, EUREKA Clusters, and H-Europe. Supervise the projects about water analytics, optical & electro-optical communication – including product design, electronic HW, embedded SW, and cloud service development. Monitor the effectiveness of research & development and improve. Execute workflow of IP Rights, QA, and cooperate with stakeholders & legals. Manage budgets and schedules in line with requirements and collaborate with peers. Establish and execute quality plans, aligning with overall business objectives. Lead researchers and innovators to develop new products and technologies. Identify new opportunities for research and innovation in the market. Represent the organization in external forums and build strategic partnerships.

Part-Time Instructor, **Information Systems Department of Okan University Graduate School**, İSTANBUL/TÜRKİYE, Oct 2023 – Feb 2024
2023-24 / Fall → Management and Organization + Scientific Research Methods

R&D Advisor to Research Centers, Nov 2022 – Nov 2023
Besides supervising the ongoing innovation projects, co-author the proposals toward funds from the Ministry, EUREKA Clusters, and H-Europe.
Latest Project: *https://itea4.org/project/nadir.html*

R&D Project Manager, **KoçSistem Inc.**,
İSTANBUL/TÜRKİYE, June 2018 – Nov 2022
Hundreds of document reviews and tens of project assessments as a member of the ITEA Steering Group. Research to solve real-life problems and develop project scopes and objectives with a forward-thinking approach, involving all relevant stakeholders, performing risk management, and ensuring technical feasibility. Detailed project plan and maintain comprehensive documentation, conforming to the resources. Coordinate internal resources and third parties/vendors for the flawless execution, and collaborate with cross-functional teams and partners with commercial awareness. Agile Project Management and ensure that all projects are delivered on time.
Hands on SW development experience parallel to management within the R&D&I projects, towards ITEA, Horizon and TÜBİTAK, consist of Federated Learning, Cognitive AI, Human Computer Interaction, Affective Computing, Computational Linguistic, Image Processing, Video Analytic, 3D Media and eXtended Reality: (i) human analytics via video stream mining for emotion recognition, age detection, gender prediction, (ii) 3D photogrammetry with depth sensing cameras and 3D point cloud data processing for XR pilots, (iii) supervised fusion of the data sources (i.e. social media, digital maps and video processing on the edge) for a target, e.g., AI for traffic density detection and assessment, (iv) multi model (e.g. video and speech analytics) affective computing for autonomous and smart digital signage – focusing on automated human behavior recognition.
Latest Projects: https://itea4.org/project/mos2s.html , https://itea4.org/project/insist.html, https://www.celticnext.eu/project-f4itech/ , https://itea4.org/project/cape.html https://itea4.org/project/sintra.html

R&D Manager, **ENTES Electronic Inc.**, İSTANBUL/TÜRKİYE, Oct 2016 – Mar 2018

With an R&D Team of 40+ persons and subcontractor companies, managed projects with the theme "**electric energy reactive power control and consumption optimization**", including product design, electronic HW, embedded SW, and cloud service development plan HW & SW development projects and supervise the run in Agile Fashion by means of Jira & Confluence. Manage budgets and schedules in line with requirements and collaborate with peers. Establish and execute quality plans, aligning with overall business objectives.

Latest Project: https://itea4.org/project/spear.html

R&D Head, **AYMED Medical Inc.**, İSTANBUL/TÜRKİYE, Mar 2016 – Oct 2016

Research and develop software for Medical Electronics R&D projects in the concepts of Urology and Remote Patient Observation by utilizing all facilities of

- C#, .NET MVC, ASP.NET Web Forms (for on-premises and cloud applications),
- SQL and Oracle DB Servers,
- Image Processing Libraries (DICOM, PACS, Eksor, OpenCV)

R&D Engineer, **SİSOFT Healthcare Inc.**, ANKARA/TÜRKİYE, Dec 2014 – May 2015

Manage Hospital Management Localization Project.

Research → Arrhythmia Detection with Wearable Devices – government-funded project.

Develop → Back-end and front-end of medical device data processing by implementing Java, JavaScript, HTML5, and Oracle DB Server technologies for the hospital management system with service-oriented architecture.

Researcher, **METU Graduate School of Informatics**, ANKARA/TÜRKİYE, Oct 2013 – July 2014

Research and implement

- ➢ image processing algorithms in .Net and MATLAB within the government-funded project – "Anomaly Detection by Image Processing of Crowded Environment Video Surveillance",
- ➢ data processing algorithms in Java and R, within the government-funded project – "Emotion Recognition by Signal Processing of Smartphone Sensor Data".

To conduct **Cognitive AI** experiments with the audience, (i) design an Android app and an MS-Windows application for wireless data collection and video annotation, (ii) develop a Windows application with C#, .NET MVC, (iii) conduct a mood recognition experiment on the audience, and (iv) analyze data and train a model.

Product Owner & Project Manager + ITC Engineer, **Ministry of Defense**, ANKARA/TÜRKİYE, Sep 1998 – Oct 2012

Completed bachelor's and Master of Science studies in Computer Engineering at Bilkent University – *ended up with a thesis about text categorization algorithms, realizing NLP and machine learning*. Since then gained knowledge and improved wisdom in various types of IT projects' development, procurement, and management:

- ➢ Land Force C4ISR projects about command and control of fire support – ADOP 2000 of ASELSAN with $3B budget, focused on: (i) wireless net centric data communication in the field, (ii) the open systems interconnection (OSI) layers enable diverse systems to communicate using standard protocols, (iii) AI for decision support to fire support planning, and (iv) cyber security.
- ➢ Turkish General Staff HQ's Strategic Level Integrated Logistics Support and Enterprise Resource Planning systems projects – implemented by TAI with a $3M budget, as focused on (i) Integrated Logistics Support for complex weapons and systems, (ii) cloud services for logistic management systems about the logistics coordination among forces, resulting in savings of $1M per year.
- ➢ Strategic level Simulation projects implemented by defense industry companies (HAVELSAN, CTECH, METEKSAN, ODTÜ-TSK MODSİMMER) – with a $12M budget, focused on training and decision support of Land, Land-Air Warfare, Logistics, and Conceptual Modelling.
- ➢ Operative-level warfare simulation training center installation, service, and maintenance with US-origin systems – JANUS and JCATS running on Linux.
- ➢ Tactical level network building, installation, improvement, and maintenance.

Managed and/or contributed hands-on experience in the projects:

- ➢ Data Analysis projects for Decision Support systems implemented on C#, C++, Java, JavaScript, PHP, and SQL.
- ➢ Advanced analytics and infrastructure solutions to support Decision Support and Simulation systems.
- ➢ Predictive models and other solutions that drive value for the organization.
- ➢ Analytics solutions and make recommendations for improvement.

Abroad Missions

Operations Centre Director,
NATO KFOR, PRIZREN/KOSOVO, Nov 2007 – June 2008
Military Observer (Operations Planning Officer),
UN Mission in Georgia (UNOMIG), June – Dec 2003
Various NATO working groups on Integrated Logistics Support and Net Centric Warfare.

References

- Prof. Dr. Halil Altay GÜVENİR, Dean of Faculty of Applied Sciences at Bilkent University, Ankara/Türkiye, guvenir@cs.bilkent.edu.tr
- Dr. Murathan KURFALI, Uppsala University, Department of Linguistics and Philology, murathan.kurfali@lingfil.uu.se
- Prof.Dr. Olcay Taner YILDIZ, Computer Engineering Department, Özyeğin University, İstanbul/Türkiye, olcay.yildiz@ozyegin.edu.tr

Languages

Turkish (native language) and English (advanced in all aspects)

Certificates

- Foundations of LLM Mastery: Fine-tuning on multi-GPUs, Vienna Scientific Cluster, (4 Hours), 2025)
- Creative Artefacts, Coursera, coursera.org/verify/D8Y8GGKQD9DQ, ((24 Hours), 2022)
- Lead Remood, Performance Evaluation by OKRs, Emeritus, (36 Hours), 2022)
- Innovation and emerging technology: Be Disruptive, Coursera, coursera .org/verify/MARPHVWHYL89, ((24 Hours), 2021)
- Intellectual Property (IP) Rights Course by SOYLU IP Rights Advisory (İstanbul), (32 Hours) June 2017)
- Software Test Course: BT Academy (İstanbul), (32 Hours) May 2017)
- Test Driven Development: BT Academy (İstanbul), (32 Hours) May 2017)
- Agile Development and Scrum Course: BT Academy (İstanbul), (32 Hours) April 2017)
- Enterprise Design Patterns Course: BT Academy (İstanbul), (32 Hours) January 2017)
- R&D Management Course: ETKİN R&D Advisory (İstanbul), ((32 Hours) January 2017)
- Applied Project Management Certificate (PMI), Defence Ministry R&D Centre (Ankara), ((1 month) December 2010)
- System Engineering Certificate, Land Forces War College, Ankara (4 months) June 1998)

Computer Skills

TensorFlow, Keras, PyTorch, OpenCV, YoLo, …
LLMs, NLP libraries, LaTeX (Overleaf), …
Python, R, Java, C++, C#, JScript, …
.Net, Eclipse, PyCharm, IntelliJ, Builder
Matlab, SPSS, R-Studio, SVN, Git

Data Science, Deep Learning, Decision Support,
Modelling & Simulation tools and libraries
IBM WebSphere, IIS, Apache
MS Access, SQL, Oracle, TOAD, SQL Server
Jira, Confluence, MS Office, MS Project, Visio

Hobbies

I walk the dog, play tennis, swim, and cycle. Rarely jogging, horse riding, and skiing.